

Algebra 1
Unit 6
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

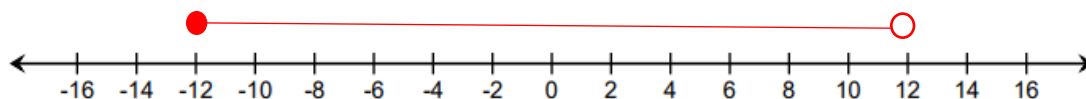
Use the following conjunctive inequality to answer questions 1-3.

$$-24 - 2x \leq -\frac{x}{3} - 4 < -2x + 16$$

Rewrite the compound inequality as two single inequalities and solve each one.

1. $(3) \quad (3) \quad (3) \quad (3)$ $-24 - 2x \leq -\frac{x}{3} - 4$ $-72 - 6x \leq -x - 12$ $+6x \quad +6x$ $-72 \leq 5x - 12$ $+12 \quad +12$ $\frac{-60}{5} \leq \frac{5x}{5} \quad -12 \leq x \quad \boxed{x \geq -12}$	2. $(3) \quad (3) \quad (3) \quad (3)$ $-\frac{x}{3} - 4 < -2x + 16$ $-x - 12 < -6x + 48$ $+6x \quad +6x$ $5x - 12 < 48$ $+12 \quad +12$ $\frac{5x}{5} < \frac{60}{5} \quad \boxed{x < 12}$
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3. Graph the solution set of the compound inequality.



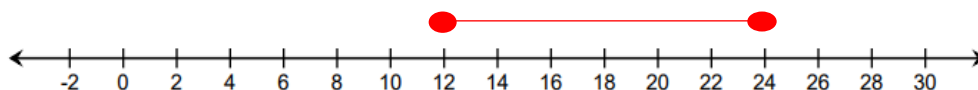
Alex just started a new sales floor job to save for college. He earns \$16.25 per hour plus a flat fee of \$55 per week. He wants to earn between \$250 and \$445 per week. The following inequality represents his earning potential:

$$250 \leq 16.25x + 55 \leq 445$$

4. Solve the inequality.

$$\begin{array}{ccc} 250 \leq 16.25x + 55 & \text{and} & 16.25x + 55 \leq 445 \\ -55 & & -55 \\ \hline \frac{195}{16.25} \leq \frac{16.25x}{16.25} & & \frac{16.25x}{16.25} \leq \frac{390}{16.25} \\ 12 \leq x & & x \leq 24 \end{array}$$

5. Graph the solution on the number line.



6. Complete the statement: Alex can work between 12 and 24 hours each week.

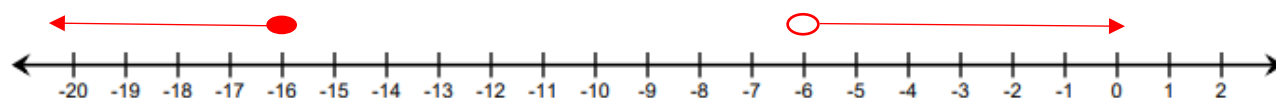
Use the disjunctive inequality shown below to answer questions 7-10:

$$-\frac{5}{3}\left(\frac{4}{3}k + \frac{8}{5}\right) > -3\frac{1}{4}k - \frac{53}{6} \text{ or } \frac{24k+14}{5} \leq -74$$

Solve each component of the compound inequality.

7. $-\frac{5}{3}\left(\frac{4}{3}k + \frac{8}{5}\right) > -3\frac{1}{4}k - \frac{53}{6}$ $\left(\frac{12}{1}\right)\left(-\frac{5}{3}\left(\frac{4}{3}k + \frac{8}{5}\right)\right) > \left(-\frac{13}{4}k - \frac{53}{6}\right)\left(\frac{12}{1}\right)$ $-20\left(\frac{4}{3}k + \frac{8}{5}\right) > -39k - 106$ $\frac{-80}{3}k - 32 > -39k - 106$ $+39k \quad +39k$ $\frac{37}{3}k - 32 > -106$ $+32 \quad +32$ $\left(\frac{3}{37}\right)\frac{37}{3}k > -74\left(\frac{3}{37}\right)$ $k > -6$	8. $\left(\frac{5}{1}\right)\left(\frac{24k+14}{5}\right) \leq -74\left(\frac{5}{1}\right)$ $24k + 14 \leq -370$ $-14 \quad -14$ $\frac{24k}{24} \leq \frac{-384}{24}$ $k \leq -16$
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9. Graph the solution sets of both inequalities on the number line.



10. Explain which of the following values is not a solution to the inequality.

-19	-9	1
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-9 is not a solution to the inequality because it does not lie in the solution set.

In addition, -9 is not shaded on the graph of the number line.